Vansh Gupta

Computer Science student

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SUMMARY

Forward-thinking Computer Science student with specialized experience in Generative AI and Multimodal system design. Demonstrated expertise in integrating Google Gemini 1.5 Flash and Pro APIs to build high-performance tools, including a medical assistant capable of complex document parsing and a multilingual desktop agent. Strong foundation in Python, asynchronous programming, and Natural Language Processing (NLP). Proven ability to optimize API latency and implement safety guardrails within AI applications.

EDUCATION

Bachelor of Technology - Computer Science Engineering — 08/2023
Visvesvaraya Technological University — Bengaluru, India

TECHNICAL SKILLS

FRAMEWORKS:
Flask

LANGUAGES:
Python, HTML, CSS, SQL

DATABASES:
SQLite, MongoDB, SQL

LIBRARIES:
Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

TOOLS:
Jupyter Notebook, Render, Deployment, VS Code, Git

ML_&_ANALYTICS:
Data Analysis, Linear Regression, Feature Engineering, EDA, Binary Classification, Data Cleaning

PROJECTS

Car Price Prediction — 01/2026 – 02/2026

- **Core Technologies:** NumPy, Pandas, Matplotlib, Linear Regression
- Built a Linear Regression model from scratch using NumPy. Focused on cost-function minimization and feature encoding to predict vehicle resale values.

Customer Churn Prediction — 02/2026 – 02/2026

- **Core Technologies:** Scikit-learn, Seaborn, EDA, Classification
- Classification pipeline identifying at-risk users based on usage patterns. Developed actionable retention strategies via deep correlation analysis.

DOC: AI-Powered Medical Conversational Assistant — 11/2025 – 12/2025

- **Core Technologies:** Python, Flask, API Integration, HTML/CSS, and Google Gemini 1.5 Flash API
- Engineered a multimodal medical chatbot using Python and Google Gemini 1.5 Flash API to provide instant health guidance and symptom analysis.
- Implemented conversational memory and context-aware logic to handle follow-up queries and maintain dialogue flow.
- Developed a document parsing system enabling the AI to analyze uploaded lab results, prescriptions, and medical records to provide personalized insights.
- Optimized API calls and backend logic to ensure low-latency response generation while maintaining professional medical tone and safety guardrails.

Nova | Gemini-Powered Multilingual AI Desktop Assistant 09/2025 – 11/2025

- **Core Technologies:** Python, Google Gemini Pro API, Tkinter, SpeechRecognition, gTTS, Pygame, PyAutoGUI.
- Developed a modular desktop assistant using Python, integrating Google Gemini 1.5 Flash for natural language understanding and real-time query resolution.
- Architected a non-blocking asynchronous system using Python's threading and queue modules, ensuring the GUI remains responsive while performing heavy speech-to-text and AI processing.
- Implemented a multilingual engine (English, Hindi, Kannada) using langdetect and gTTS, enabling the assistant to identify user language and respond with localized speech synthesis.
- Integrated OS-level automation features using PyAutoGUI and subprocess to allow the assistant to control system hardware, launch applications, and perform web searches.
- Designed a modern, dark-themed GUI using Custom Tkinter elements, featuring real-time status monitoring and a scrolled conversation log.

Full-Stack Developer Portfolio & Contact Management System 04/2026 – 05/2026

- **Core Technologies:** React.js, Tailwind CSS, FastAPI, Starlette, PostgreSQL, SQL, Neon (Cloud SQL), Vercel (Frontend Deployment)
- Architected a full-stack portfolio using React and FastAPI to showcase AI/ML projects. Built a custom contact engine with a PostgreSQL cloud database (Neon) for persistent lead management, replacing third-party form tools with a secure, scalable REST API.